Confidence interval for difference of two means, dependent samples

Weight loss example, kg

Background The 365 team has developed a diet and an exercise program for losing weight. It seems that it works like a charm. However, you are interested in how much weight are you likely to lose.
You have a sample of 10 people who have already completed the 12-week program.

Task 1 Calculate the mean and standard deviation of the dataset

Task 2 Determine the appropriate statistic to use

Task 3 Calculate the 95% confidence interval

Task 4 Interpret the result

Optional You can try to calculate the 90% and 99% confidence intervals to see the difference. There is no solution provided for these cases.

Solution:

Subject	Weight before (kg)	Weight after (kg)	Difference
1	103.68	92.87	-10.81
2	110.68	101.58	-9.10
3	119.05	105.66	-13.39
4	101.75	96.18	-5.57
5	91.69	86.97	-4.72
6	112.03	105.90	-6.13
7	88.84	80.56	-8.28
8	105.18	97.00	-8.18
9	110.37	99.27	-11.10
10	120.99	107.44	-13.55

Task 1: **Mean** -9.08
St. deviation 3.11

Task 2: Population variance is unknown
We have a small sample
We assume that the population is normally distributed
The appropriate statistic to use is the t-statistic

Note that the solution is exactly the same no matter the unit of measurement

Task 3:

95% CI, $t_{9,0.025}$ 2.26

T	CI low	CI high
95%	-11.31	-6.86

Task 4: You are 95% confident that you will lose between 11.31 and 6.86 kg, given that you follow the program as strict as the sample